
Constraint Satisfaction

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course



WHAT ARE CONSTRAINT SATISFACTION PROBLEMS?

Special kind of search problems.

- N variables
- Values from domain D
- assignment satisfies constraints

States: partial assignment

Goal Test: satisfies all constraints

Successor Function: assign an unassigned variable

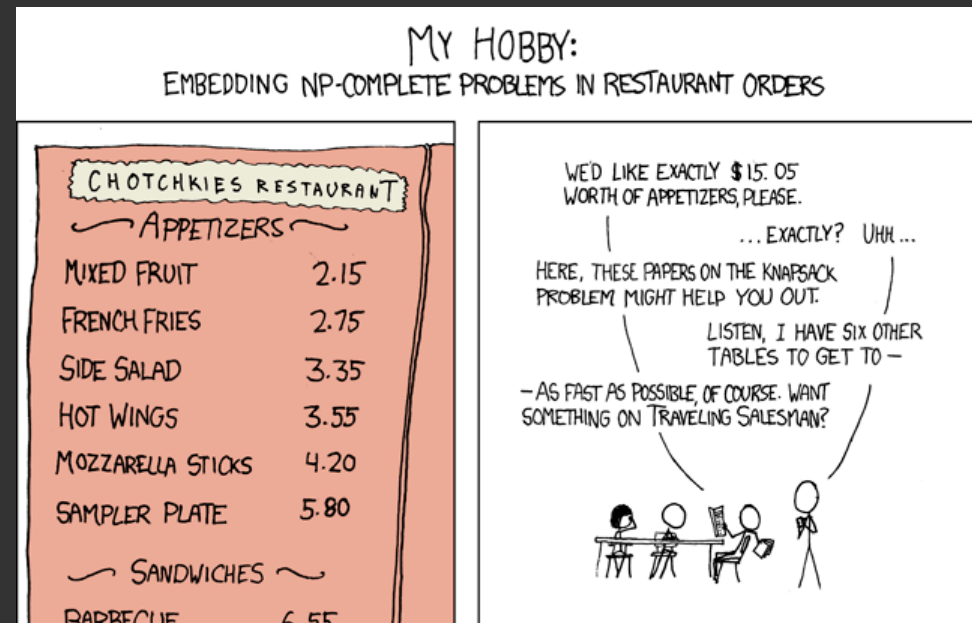


TODAY

Constraint Satisfaction

Reading

- Today's Lecture: RN Chapter 6



XKCD

WHAT IS SEARCH FOR?



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Given assumptions about the world: a single agent, deterministic actions, fully observed state, discrete state space



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- Planning: sequences of actions
 - The path to the goal is the important thing
 - Paths have various costs, depths
 - Heuristics give problem-specific guidance
- Identification: assignments to variables
 - The goal itself is important, not the path
 - All paths at the same depth (for some formulations)
 - CSPs are a specialized class of identification problems



CONSTRAINT SATISFACTION PROBLEMS



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- Goal test can be any function over states
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Constraint satisfaction problems (CSPs):

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- State is defined by variables $\mathbf{X}_i$ with values from a domain D (sometimes D depends on i)
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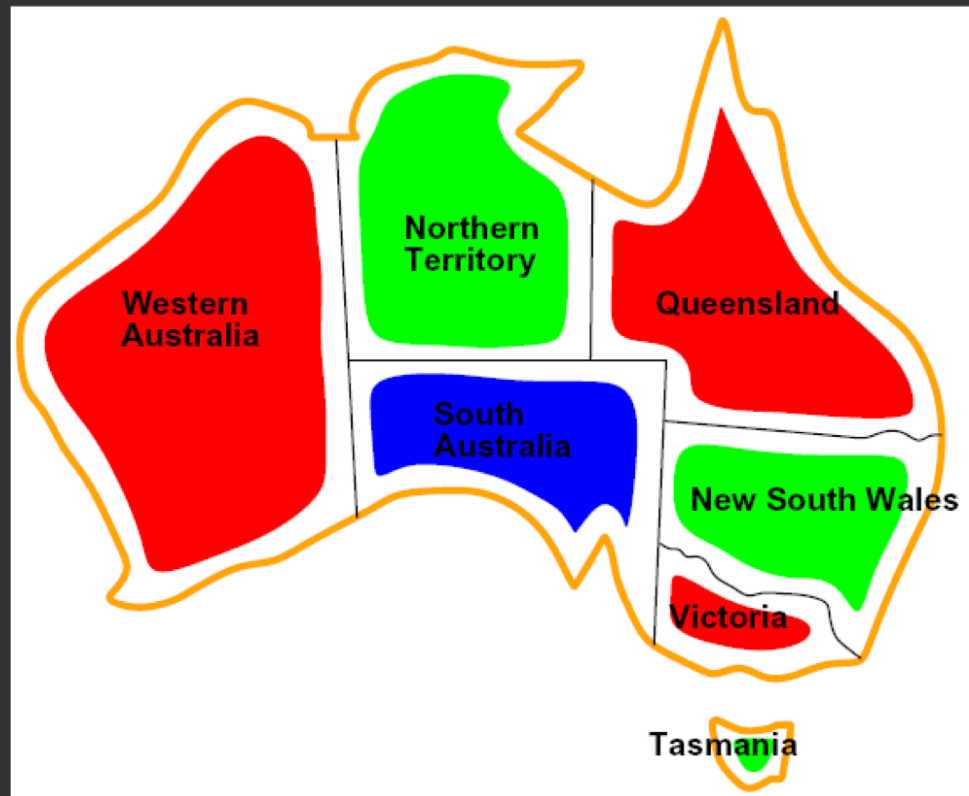
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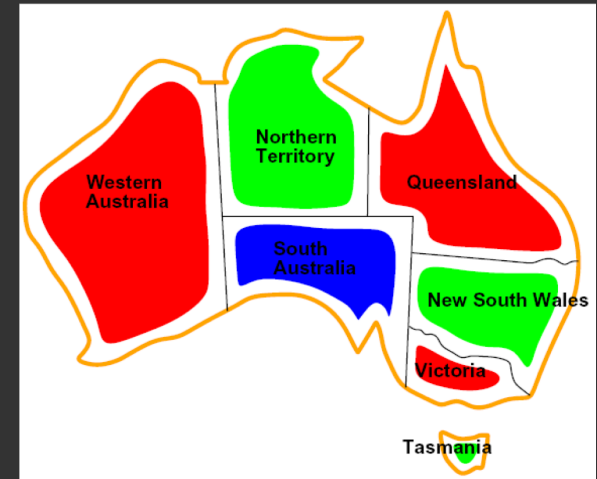
Allows useful general-purpose algorithms with more power than standard search algorithms



CSP EXAMPLE

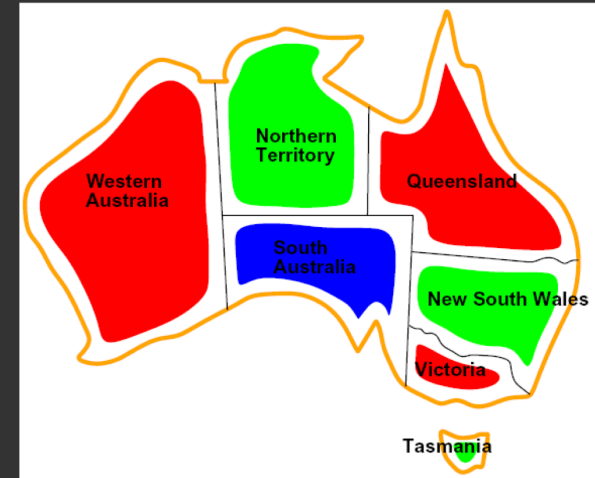


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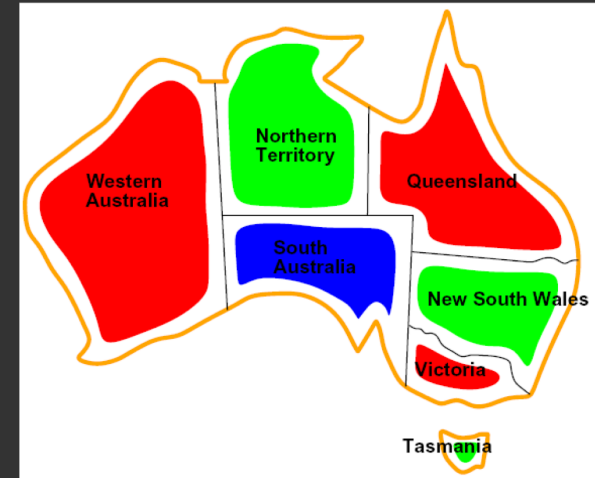
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EXAMPLE: MAP COLORING

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Domains: $D = \{red, green, blue\}$

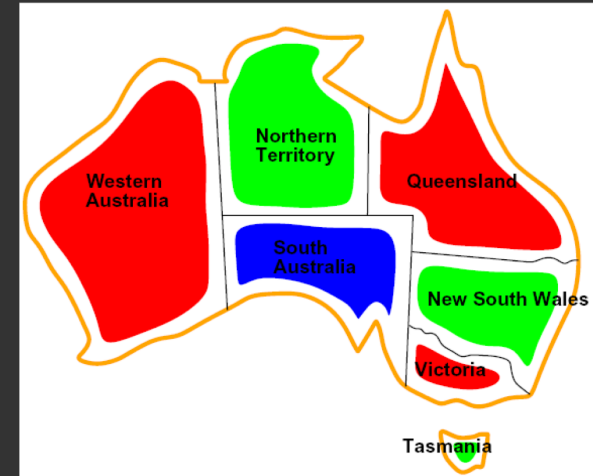


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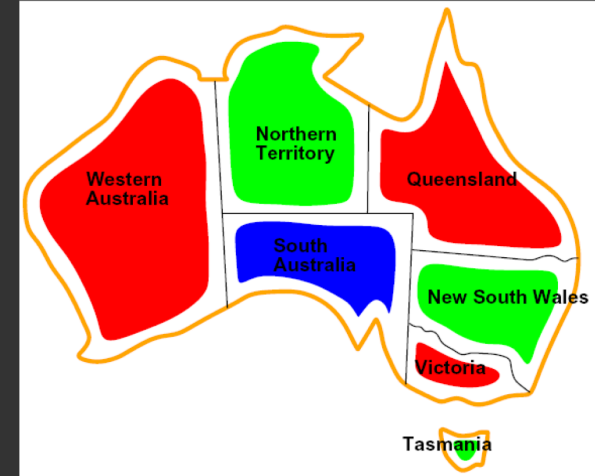
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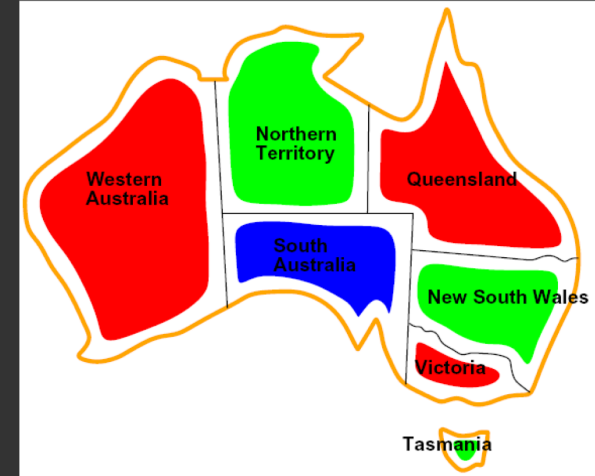
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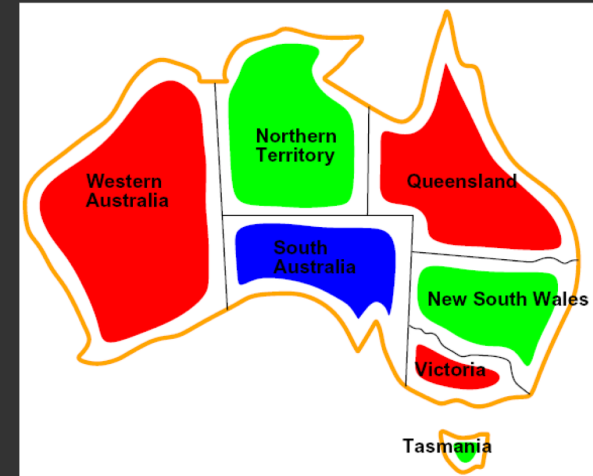
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General-purpose CSP algorithms use the graph structure to speed up search. E.g., Tasmania is an independent subproblem.

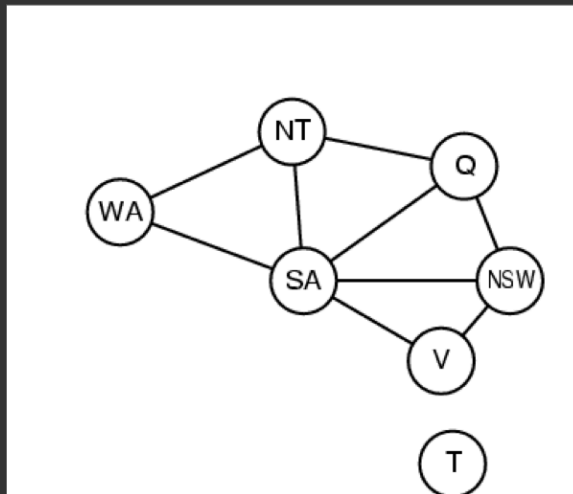


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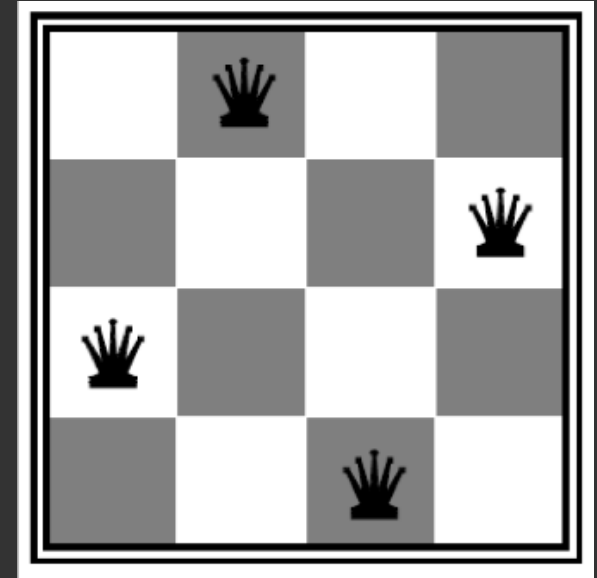
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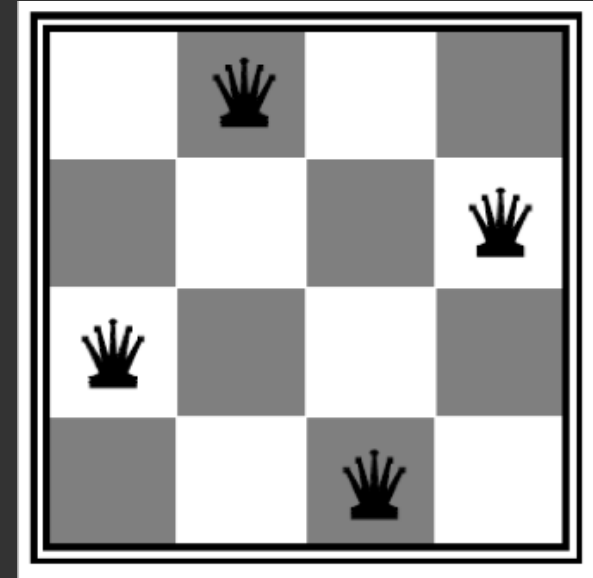


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Formulation 1:



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Formulation 1:

- Variables: X_{ij}
- Domains: $\{0, 1\}$
- Constraints:

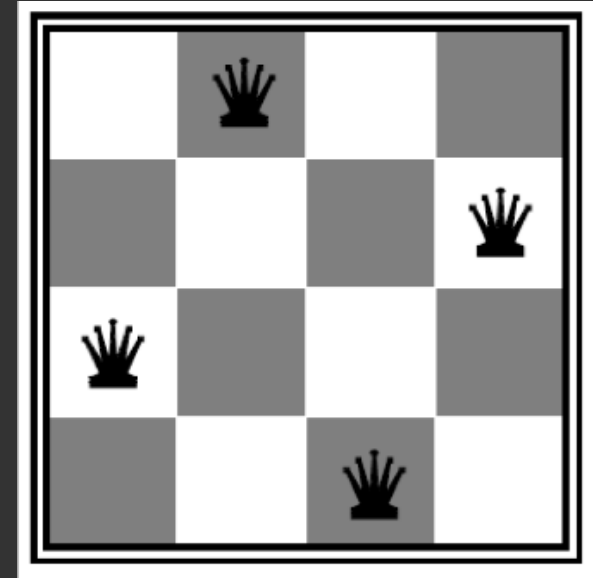
$$\forall i, j, k \quad (X_{ij}, X_{ik}) \in (0, 0), (0, 1), (1, 0)$$

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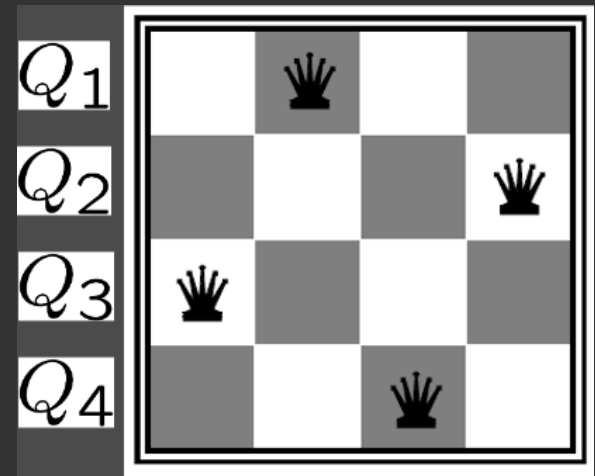
$$\forall i, j, k \quad (X_{ij}, X_{i+k, j+k}) \in (0, 0), (0, 1), (1, 0)$$

$$\forall i, j, k \quad (X_{ij}, X_{i+k, j-k}) \in (0, 0), (0, 1), (1, 0)$$

$$\sum_{ij} X_{ij} = N$$

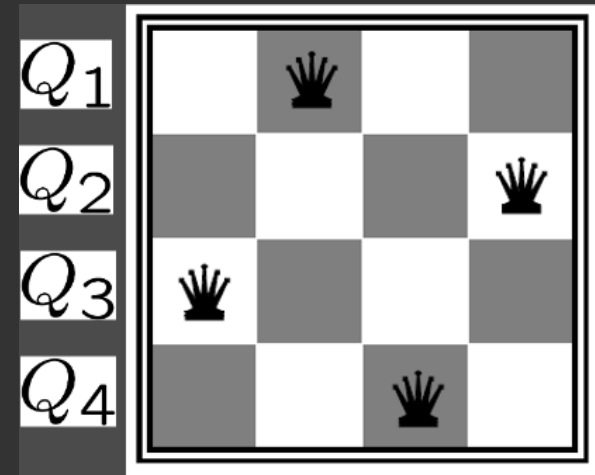


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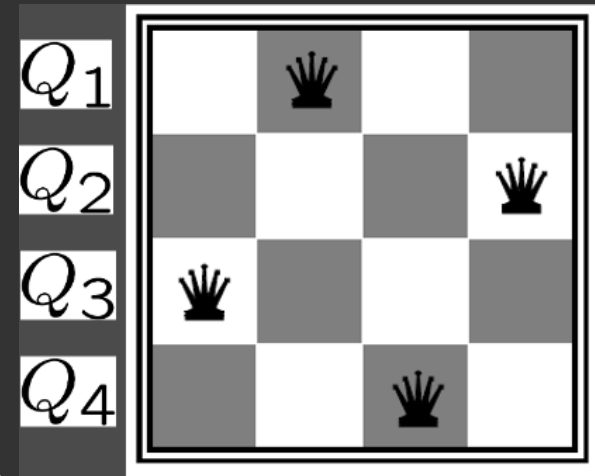
Formulation 2:



EXAMPLE: N QUEENS

Formulation 2:

- Variables: Q_k
- Domains: $1, 2, 3, \dots, N$
- Constraints:
 - Implicit: $\forall(i, j)$ non – threatening (Q_i, Q_j)
 - Explicit: $(Q_1, Q_2) \in \{(1, 3), (1, 4), \dots\}$



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Discrete Variables



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Continuous variables

- E.g., start/end times for Hubble Telescope observations
- Linear constraints solvable in polynomial time by LP methods



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Preferences (soft constraints):

- E.g., red is better than green
- Often representable by a cost for each variable assignment
- Leads to constrained optimization problems



REAL-WORLD CSPS

Scheduling problems: e.g., when can we all meet?

Timetabling problems: e.g., which class is offered when and where?

Assignment problems: e.g., who teaches what class

Hardware configuration

Transportation scheduling

Factory scheduling

Circuit layout

Fault diagnosis

...



SOLVING CSPS



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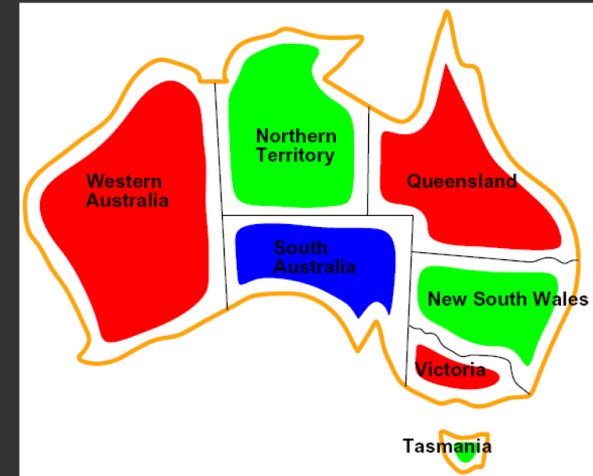
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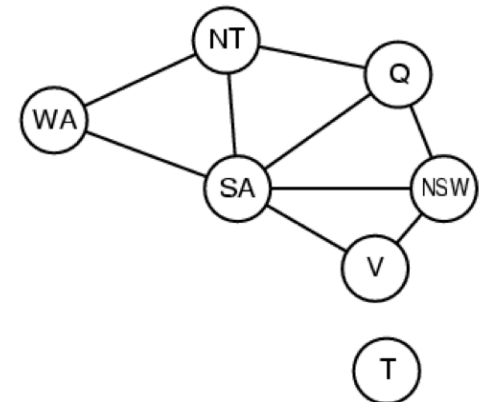
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Start with straightforward search and then improve.



SEARCH METHODS

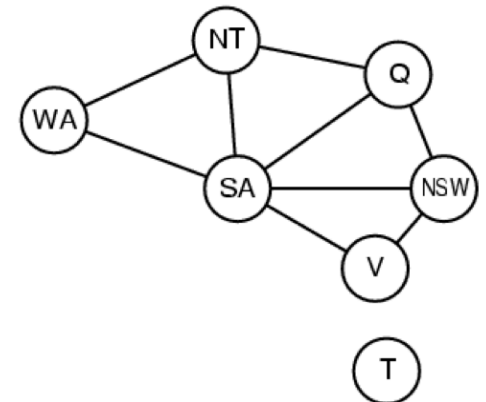
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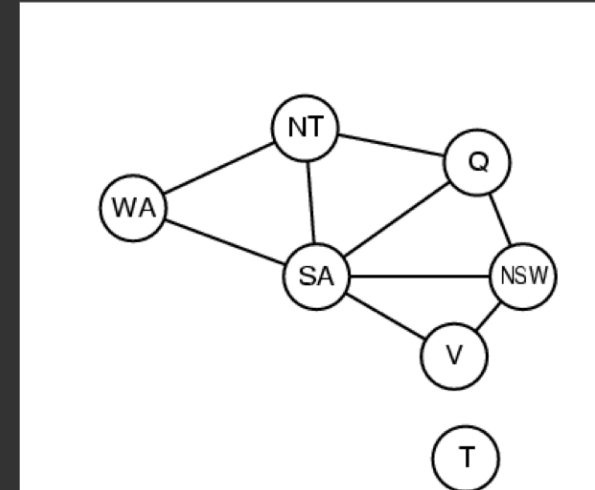


SEARCH METHODS

What would BFS do?

What would DFS do?

What problems does naïve search have?



Q & A



XKCD

